



Solana Track · Hackathon Guide

Building on Solana

A Guide for Hackathon Participants

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What's in this guide



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Why Solana: a new financial layer



Blockchain is a new financial layer. Solana is already operating as one.

200M+

monthly stablecoin
transactions

5M+

monthly active stablecoin
addresses

\$17B+

stablecoin supply

\$650B+

monthly settlement volume

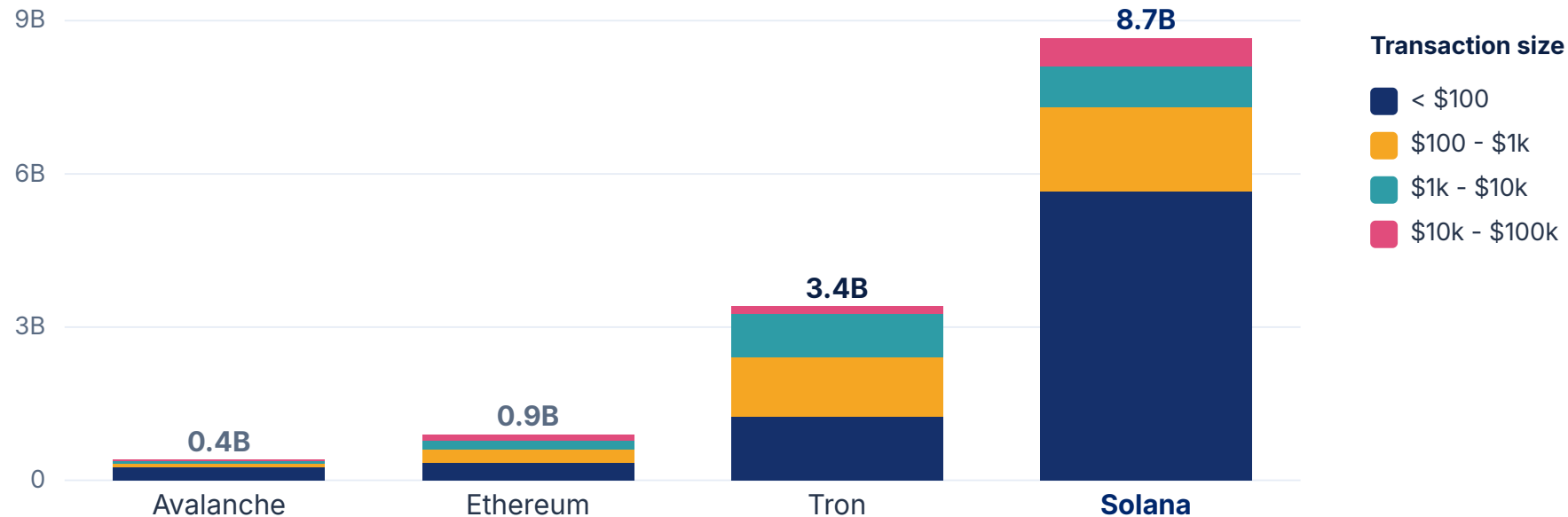
One cross-border transfer	SWIFT wire	A transfer on Solana
Time to arrive	1–5 business days	Under 1 second (~400 ms)
Fee per transfer	\$15–\$50	Less than \$0.001 (a fraction of a cent)
Intermediaries	1–3 correspondent banks	None: direct, wallet to wallet
Availability	Weekday banking hours, cut-offs & holidays	24 / 7 / 365 , always on
Tracking	Limited, opaque hops	Real-time and public , end to end

Solana mainnet (slot time ~400 ms, network fees); stablecoin metrics via Visa Onchain Analytics · Allium; SWIFT figures from industry data

Solana already carries real, everyday money



The data on where stablecoins actually move is striking: the activity is overwhelmingly retail, and it happens on Solana.



- **8.7B+ stablecoin transactions on Solana**, more than Ethereum, Tron & Avalanche combined
- **~2 in 3 are under \$100**: real retail payments, not just large-value settlement. Tron and Ethereum skew to larger transfers; Solana is where everyday money actually moves.

Source: Visa Onchain Analytics (visaonchainanalytics.com/transactions); Solana ranks #1 even with all chains selected, not just those shown.

Global financial institutions already settle on Solana



Three of the most regulated names in payments run production stablecoin flows on Solana today, spanning bank settlement, global remittance, and consumer payments.



BANK-TO-BANK SETTLEMENT

Launched **USDC settlement on Solana** for U.S. banks (Cross River Bank, Lead Bank).

Over **\$3.5B annualized** stablecoin settlement run-rate.



GLOBAL REMITTANCE

Launched **USDPT**, its USD stablecoin, on Solana (May 2026).

Issued by **Anchorage Digital Bank**, a federally regulated US crypto bank.



CONSUMER PAYMENTS

PYUSD issued natively on Solana (Paxos, NYDFS-regulated).

Used for cross-border remittance and creator payouts.

What's new: Alpenglow



IN SIMPLE TERMS

Solana is run by thousands of computers (validators). They must agree on which transactions happened and in what order. The rules for reaching that agreement are called the *consensus protocol*. Alpenglow is a brand-new, redesigned set of those rules. The result: the network says "your transaction is final and cannot be reversed" far sooner than before.

Whitepaper estimate: ~150 ms → Measured on Testnet: **54 ms median**

- **What "finality" means:** the moment a transaction is confirmed and can no longer be undone. Faster finality = payments, trades and games feel instant.
- **Rollout:** Testnet  → **Devnet**  (Oct 1) → Mainnet: coming soon
- **For app developers:** nothing to change. Your existing code keeps working and simply gets faster confirmations.
- **Learn more:** "What's Alpenglow?" whiteboard session with Jacob Creech and Roger Wattenhofer: youtu.be/2RINav7GiUY

What's new: V1 transactions



IN SIMPLE TERMS

A Solana transaction is like an envelope: it holds the instructions you want to run, the accounts they touch, and the signatures. The envelope used to have a hard limit of **1232 bytes** (about one network packet). **V1 is a new envelope format with a 4096-byte limit**, about 3x bigger, and it is live on mainnet today.

Why the old limit hurt

- Complex DeFi or game actions had to be split across several transactions
- Deploying a program meant hundreds of small upload transactions
- Heavy cryptography (signatures, proofs) did not fit

What V1 enables

- More instructions and accounts in a single atomic transaction
- Faster deploys: one developer went from **6 min 50 s** to **1 min 44 s** (52 txs) for the same program
- New programs built on it, e.g. the ed25519 programmatic signer

WHAT YOU NEED TO DO

Update your SDK (Web3.js v3 / Kit, Rust SDK) and CLI, then opt in when building a transaction. Legacy and v0 transactions keep working, so nothing breaks if you wait.

Why build on Solana now



Solana is getting faster, gaining new capabilities, and being optimized every week, on the way to becoming the home of **Internet Capital Markets**.

- **Faster:** Alpenglow brings finality down to tens of milliseconds
- **More capable:** V1 transactions, sBPFv3 programs, new SDKs (Web3.js v3, Kit, Anchor 1.x)
- **Optimized:** two independent validator clients (Agave, Firedancer) competing on performance, with weekly improvements shipped in the Solana Changelog

WHAT WE ARE LOOKING FOR

Projects that are **only possible on Solana**, or that make Solana's strengths obvious: sub-second finality, low fees, high throughput, and composability with existing on-chain programs. If your idea would work the same on any chain, ask yourself what Solana lets you do that others cannot.

Solana Track: build on the fastest chain



Total prize pool: **\$10,000**

\$5,000

1st place

\$3,000

2nd place

\$2,000

3rd place

WHAT COUNTS AS "BUILDING ON SOLANA"

The project must interact with the Solana blockchain, either through a program you deploy or by meaningfully integrating existing Solana programs. A frontend that only reads on-chain data does not qualify.

Qualification checklist



- ☐ The project must interact with the Solana blockchain, either through a program you deploy or by meaningfully integrating existing Solana programs. A frontend that only reads on-chain data does not qualify.
- ☐ The application must be functional on either Devnet or Mainnet Beta. If you have deployed your own program, please specify the Program ID and the cluster it is deployed on.
- ☐ Include at least one example transaction link from Solana Explorer or Solscan.
- ☐ Code must be written during the hackathon. Any pre-existing work must be disclosed in the README.
- ☐ Public GitHub repository, or judge access granted.
- ☐ A working demo that a judge can run end to end without assistance.

Judging criteria



Criterion	Weight	What the judges look for
Technical Execution on Solana	30%	How well the project uses Solana. Projects whose core logic lives on-chain score higher than those that merely touch the chain.
Innovation & Originality	20%	Whether this is a new idea or a clearly better take on an existing one.
Product & User Experience	20%	Whether it works and is pleasant to use, including onboarding and wallet flow.
Real-World Impact & Viability	15%	Who needs this, and whether there is a plausible path to users after the hackathon.
Demo & Presentation	15%	How clearly the team shows what they built. A live demo counts for more than slides.

Getting started: docs and tools



Resource	What it is for
Solana Docs: Quick Start	Official Quick Start guide
Solana Developer Hub	Central hub for all dev resources
Solana Playground	Write, deploy & test programs in the browser (zero setup)
Solana Wallet Adapter	Wallet connection library
Program Examples	Solana Program (Smart Contract) Examples: production-grade examples (NFTs, DeFi, governance)
Solana Stack Exchange	Technical Q&A

Getting test SOL: airdrops have changed



HEADS UP

`solana airdrop` on the CLI **no longer works**. Test SOL for Devnet and Testnet is only available through the web faucet: faucet.solana.com. The CLI will now point you there.

- **Connect your GitHub account** on the faucet to receive a larger amount per request
- **Develop on Localnet whenever possible:** run `solana-test-validator` on your machine. It gives you unlimited SOL, instant resets, and no rate limits. Use Devnet only for integration testing and your final demo.
- **Running short on Devnet SOL?** Ask in the **Token2049 hackathon Telegram channel** and the Solana team will help you out

Learn and get inspired



Resource	What it is for
Solana Developer Bootcamp 2026 + bootcamp hub page	Rust, Anchor, PDAs, full-stack Web3 apps (Full Course)
Colosseum Project Explorer	Browse all Solana's official global hackathons' past submissions, filter by track or country, toggle to show only winners
Colosseum Copilot	AI research skill for your coding assistant (Claude Code, Codex). Vet your idea against 5,400+ past submissions and live ecosystem data before you build.
Awesome Solana AI	Curated list of AI coding skills and tools for Solana development (Anchor, Metaplex, Helius skills for Claude Code, etc.), maintained by the Solana Foundation.

Who to ask: your on-site team



Name	Role	Email
Andre Correia Head of Developer Relations	Judge & Mentor	andre.correia@solana.org
Chaerin Kim Developer Relations	Mentor	chaerin.kim@solana.org
Mike Ma Developer Relations	Mentor	mike.ma@solana.org